



Wildfire Detection Node

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Abstract

This project presents a low-power wildfire detection sensor node designed for early fire detection in Pacific Northwest forests. The system uses MLX90640 infrared thermal array sensors to monitor environmental temperature patterns and detect abnormal heat signatures. An STM32-based MCU processes the thermal data, estimates battery level, and transmits alerts using LoRa wireless communication. The node is designed for long-term outdoor deployment using a compact battery, scheduled operational cycles, and a weather-resistant enclosure with tree mounting features. Our custom PCB integrates the sensing, processing, communication, and power-management hardware into a compact embedded system. Our project is affordable, scalable, reliable, and open-source.

Background

Wildfires are an increasing threat to Pacific Northwest forests, communities, and ecosystems. Early detection is important because small ignition events are much easier to contain before they grow into large fires.

The Whatcom Million Trees Project requested an affordable and open-source wildfire detection system that could support long-term deployment. Many modern commercial systems are expensive to deploy and maintain, which limits availability for smaller organizations, rural communities, and environmental groups.

Our project addresses this need by developing a low-cost, low-power sensor node that can be deployed in forested areas to detect abnormal heat signatures and wirelessly report possible fire threats.

References

[1] Dryad Networks GmbH, "Wildfire Detection Technology," Dryad Networks, Berlin, Germany. [Online]. Available: <https://www.dryad.net>

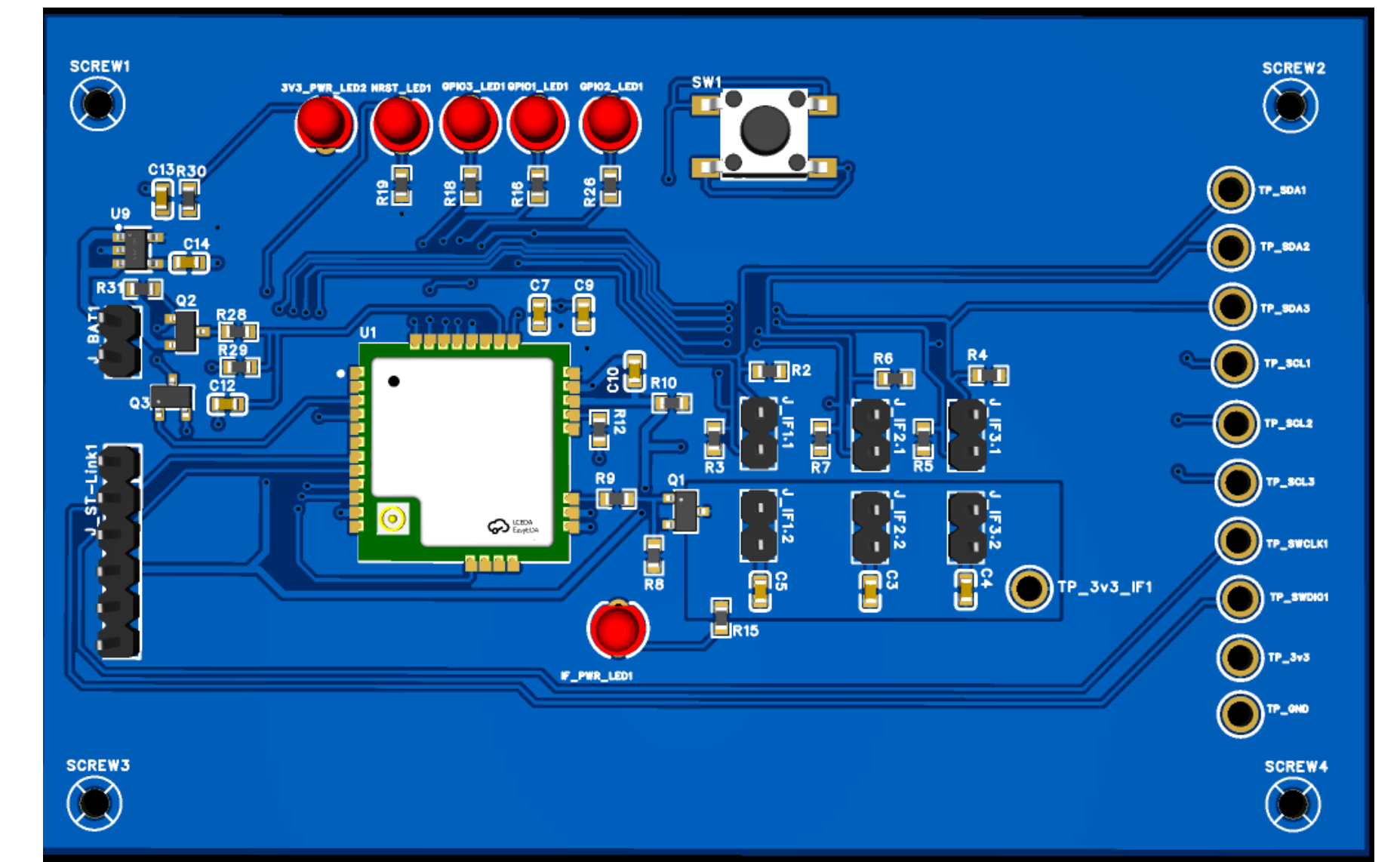
[2] Dryad Networks GmbH, "Early Wildfire Detection Using Gas Sensors in Forest Environments," White Paper, 2022.

Methods and Materials

The system uses a low-power wireless sensor node built around a RAK3172 module, which combines an STM32WLE5 microcontroller with an integrated Semtech LoRa radio. A MLX90640 infrared thermal array monitors the environment for elevated heat signatures, while an ADC measures battery voltage for system health tracking. The node is powered by a 3.6 V lithium thionyl chloride battery and uses MOSFET-controlled power switching to disable the infrared sensor plane and ADC during sleep.

To reduce current consumption, the node follows a duty-cycled operating sequence. The microcontroller remains in a low-power mode between measurement cycles. After waking, the node powers the infrared sensor, collects thermal data, measures battery voltage, classifies the fire threat level, transmits a LoRa packet, and returns to sleep.

The embedded software classifies sensor readings into threat levels representing no threat, fire candidate, confirmed fire alert, and infrared sensor maintenance fault. LoRa packets include the node ID, battery voltage, maximum detected temperature, and threat level. A gateway computer receives and decodes these packets using a Python script, logs each alert with timestamp, node ID, RSSI, and threat level, and sends SMS notifications for confirmed fire alerts or maintenance faults.



Results

The prototype demonstrated the core architecture of a wildfire sensing node, including the custom PCB, MLX90640 infrared sensor, battery voltage sensing circuit, LoRa radio, and MOSFET-controlled sensor power switching. The gateway software successfully received LoRa packets, decoded packet contents, logged alert data with timestamp, node ID, RSSI, and threat level, and generated SMS notifications for fire alerts and maintenance faults.

Subsystem testing confirmed that the node can collect thermal data, estimate maximum temperature, measure battery voltage, classify basic fire conditions, and format alert packets. Fire classification worked under controlled bench testing, especially with small flames placed directly in front of the MLX90640 sensor, but additional tuning is needed for reliable outdoor detection at longer distances.

Power management was partially demonstrated through sensor power gating and STOP2 sleep implementation. However, sleep current remained above the target value, and STOP2 behavior was not reliable enough to meet the long-duration battery-life goal. LoRa communication was functional in controlled tests but showed reliability issues during longer-range and forested testing.

Overall, the prototype validates the system concept while identifying power optimization, LoRa reliability, and improved fire classification as the main areas for future work.

Future Direction

By reducing the cost of wildfire monitoring, the system has the potential to make early detection technology more accessible to nonprofit organizations, rural communities, and land managers.

For full-scale field deployment, the prototypes current 3D printed enclosure will have to be converted to an IP-rated material. All openings will need to be sealed with IP-rated gasket sealant. These measures will ensure the longevity of the system in the Pacific Northwest environments.

As hardware such as the LoRa module and infrared thermal array sensors improve and older models become less expensive, the system can be upgraded without sacrificing affordability.

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